

Mid-term examination on XML

Christian Rinderknecht

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1 Queues

A *queue* is a stack where the items are popped at the **bottom**, not at the top. Pushing an item in a queue is called *enqueueing*, whereas the special pop is called *dequeueing*.

ENQUEUE $\rightarrow$

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	
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 $\rightarrow$ DEQUEUE

PUSH, POP $\leftrightarrow$

<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
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We want to implement queues with two stacks:

ENQUEUE $\rightarrow$

<i>a</i>	<i>b</i>	<i>c</i>
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<i>d</i>	<i>e</i>
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 $\rightarrow$ DEQUEUE

So ENQUEUE is PUSH on the first stack and DEQUEUE is POP on the second. If the second stack is empty, we swap the stacks and reverse the (new) second:

ENQUEUE $\rightarrow$

<i>a</i>	<i>b</i>	<i>c</i>
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 $\rightarrow$ DEQUEUE ???

ENQUEUE $\rightarrow$

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<i>a</i>	<i>b</i>	<i>c</i>
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 $\rightarrow$ DEQUEUE

Data. Let $\text{QUEUE}(S, T)$ be the queue made of the stacks S and T ; let $\text{PUSH}(e, S)$ be the stack S with item e on top; let EMPTY be the empty stack.

Functions. Let $\text{ENQUEUE}(e, Q)$ rewrite to the queue Q where item e is added; let $\text{DEQUEUE}(Q)$ rewrite to the pair (e, Q') where e is the item on the head of queue Q and Q' is the remaining queue.

Question. Define $\text{ENQUEUE}(e, Q)$ and $\text{DEQUEUE}(Q)$, where Q is a queue.

2 Stacks

Question. Define $\text{BOTTOM}(S)$ be the pair (e, T) where e is the item at the bottom of stack S and T is S without e . There is a direct way to make queues (that is, without using stacks). We need here an empty queue, noted Q_0 , but **no** QUEUE .

Example. The queue containing only item e is $\text{ENQUEUE}(e, Q_0)$. The queue containing item e and then f is $\text{ENQUEUE}(f, \text{ENQUEUE}(e, Q_0))$. We have

$$\text{DEQUEUE}(\text{ENQUEUE}(f, \text{ENQUEUE}(e, Q_0))) \rightarrow \cdots \rightarrow (\text{ENQUEUE}(f, Q_0), e)$$

Question. Give the rewrite rules defining DEQUEUE .